

The theory of the consumer

The central assumption in the theory of the consumer is optimization: given the feasible set of consumption bundles, the bundle chosen is the one the consumer prefers. The purpose of the theory is first to characterize the bundle of goods which will be chosen, and second to predict how the optimal choice will change in response to changes in the feasible set.

In analysing the consumer's optimal choice, we proceed in three steps. We first construct a model of the consumer's preferences, which allows us to specify certain properties of the consumer's ranking of consumption bundles. We then examine how the prices of commodities, in conjunction with the consumer's income (or initial endowment of commodities in a more general model), determine the feasible set of consumption bundles. Finally, by applying the model of the consumer's preference ordering to the feasible set, we are able to determine the characteristics of the optimal choice.

A. The preference ordering

A consumption bundle is denoted by a vector:

$$x = (x_1, x_2, \dots, x_n)$$

where x_i , $i = 1, 2, \dots, n$, is the amount of the i th good in the bundle. Each x_i is assumed to be non-negative – the consumer can consume only zero or a positive quantity of each good – and also is taken to be perfectly divisible – goods do not come in lumpy discrete amounts.

The meaning of the terms 'preference' and 'indifference' is taken as understood; we take it for granted that everyone knows what is meant by the statement, 'I prefer this to that', or, 'I am indifferent between this and that'. We assume that the consumer can make statements such as, 'I prefer consumption bundle x' to x'' ', or, 'I am indifferent between x' and x'' '. More formally, we introduce the symbol $\succeq$ which is read 'is preferred or indifferent to', or 'is at least as good as', or 'is no worse than', so that $x' \succeq x''$ means that the consumer regards x' as at least as good as x'' . This symbol is called the *preference–indifference relation*.

The consumer ranks the consumption bundles in the feasible set in order of preference, and chooses the one which is highest in the ranking. This preference ranking can be thought of as being arrived at by repeated application of the preference–indifference relation to successive pairs of consumption bundles. For the purpose of our theory, we want the preference ranking to have certain properties, which give it a particular, useful structure. We build these properties up by making a number of assumptions, first about the preference–indifference relation itself, and then about some aspects of the preference ranking to which it gives rise.

As a preliminary, suppose the consumer told us that

$$x' \succeq x'' \quad \text{and} \quad x'' \succeq x'$$

in words, 'x' is preferred or indifferent to x'', and 'x'' is preferred or indifferent to x'. Since we would regard him as talking nonsense – violating the meaning of the word 'preferred' – if he told us that x' is preferred to x'' *and* x'' is preferred to x', this must mean that x' is indifferent to x''. We write 'x' is indifferent to x'', as $x' \sim x''$. Suppose, alternatively, the consumer told us that:

$$x' \succeq x'' \quad \text{and not} \quad x'' \succeq x'$$

This must mean that x' is preferred to x'' and this is written $x' > x''$. Thus we have as implications of the meaning of the preference–indifference relation:

- (a) $x' \succeq x''$ and $x'' \succeq x'$ implies $x' \sim x''$
- (b) $x' \succeq x''$ and *not* $x'' \succeq x'$ implies $x' > x''$.

We can now proceed to the assumptions which give the desired properties to the consumer's preference ordering.

Assumption 1. Completeness. For any pair of bundles x' and x'' , either $x' \succeq x''$ or $x'' \succeq x'$ (or both).

This assumption says in effect that the consumer is able to express a preference or indifference between any pair of consumption bundles however alike or unlike they may be. This ensures that there are no 'holes' in the preference ordering. It also implies that, given some bundle x' , every other bundle can be put into one of three sets:

1. the set of bundles preferred or indifferent to x' : the 'better set' for x' ;
2. the set of bundles indifferent to x' : the 'indifference set' of x' ;
3. the set of bundles to which x' is preferred or indifferent: the 'worse set' for x' .

These sets, and especially set 2, play an important part in what follows.

Assumption 2. Transitivity. For any three bundles x' , x'' , x''' , if $x' \succeq x''$ and $x'' \succeq x'''$ then $x' \succeq x'''$.

This is a consistency requirement on the consumer. Given the first two statements, if the third did not hold, so that $x''' > x'$, we would feel there was an inconsistency in preferences. The assumption has an important implication for the 'indifference sets' just defined, in that it implies that no bundle can belong to more than one such set. For suppose that $x' \sim x''$, so that x'' belongs to the indifference set of x' ; and also that $x'' \sim x'''$, so x'' belongs to the indifference set of x''' . If $x' \sim x'''$, then there is no problem, since all three bundles are in the same indifference set. But suppose $x''' > x'$. Then x'' must be in two indifference sets, that of x' and that of x''' . But then we have

$$x' \sim x'' \quad \text{and} \quad x'' \sim x''' \quad \text{but} \quad x''' > x'$$

which violates the assumption of transitivity. Thus given this assumption, no bundle can belong to more than one indifference set: *the transitivity assumption implies that indifference sets have no intersection.*

Assumption 3. Reflexivity. $x' \succeq x'$.

In words, any bundle is preferred or indifferent to itself. Since we can interchange the two sides of the relation, the assumption has the implication that a bundle is indifferent to itself, which seems trivially true. However, its implication is less trivial: it ensures that every bundle belongs to at least one indifference set, namely that containing itself, if nothing else.

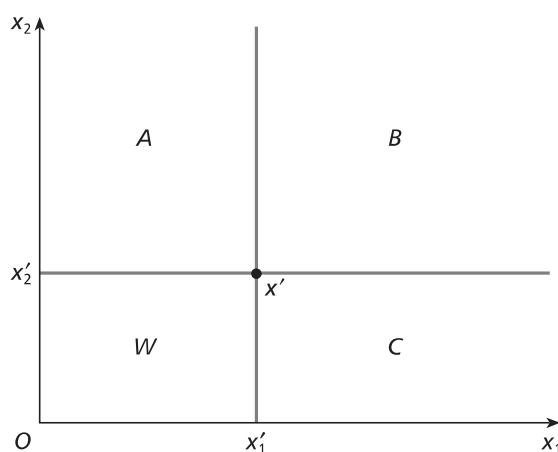
These three properties of the preference–indifference relation imply that every bundle (completeness) can be put into one indifference set (reflexivity) and no more than one indifference set (transitivity). Thus we can *partition* any given set of consumption bundles, by use of the relation, into non-intersecting indifference sets, which provide us with a useful way of representing a particular preference ordering. The indifference sets can be ranked in order of preference on the basis of the ranking of the bundles they contain. The following assumptions we make about the consumer's preferences are chiefly designed to give these sets a particular structure.

Assumption 4. Non-satiation. A consumption bundle x' will be preferred to x'' if x' contains more of at least one good and no less of any other, i.e. if $x' \succ x''$.

This assumption establishes a relationship between the quantities of goods in a bundle and its place in the preference ordering – the more of each good it contains the better. Moreover, this is held to be true however large the amounts of the goods in the bundle, hence the term 'non-satiation' – the consumer is assumed never to be satiated with goods. This assumption is much stronger than we need to make in two respects. It first implies that none of the goods is in fact a 'bad', a commodity such as garbage or aircraft noise which one would prefer to have less of. Second, it assumes that the consumer is never satiated in *any* good. We could generalize by allowing some goods to be bads, and by assuming non-satiation only in at least one good, without changing anything of significance in the results of the theory. For simplicity, however, we adopt the stronger assumption here.

The non-satiation assumption has two important consequences for the nature of indifference sets, which are best expressed geometrically. In Fig. 2.1, x_1 and x_2 are goods, and $x' = (x'_1, x'_2)$ is a consumption bundle. Because of assumption 4, all

Figure 2.1



bundles in the area B (including the boundaries, except for x' itself) must be preferred to x' , and all points in the area W (again including the boundaries except for x') must be inferior to x' . The first consequence of the assumption is that points in the indifference set for x' (if there *are* any besides x') must lie in areas A and C . In other words, if we imagine moving between bundles in the indifference set, we can only do so by *substituting* or *trading off* the goods – giving more of one good must require taking away some of the other good in order to stay within the indifference set. The second consequence is that an indifference set is never ‘wider’ than a single point – its geometric representation can never be an area or band, though it may be a single point, an unconnected set of points or a curve. For suppose x' was contained in an indifference set which was a band. Then some bundles indifferent to it must lie in areas B and W , which violates assumption 4. Thus the assumption implies that an indifference set cannot be thick at point x' , or, by extending the argument, at any point contained in it.

None of the assumptions we have made so far, however, implies that there must be more than one point in an indifference set, or, if there is more than one point, that these make up a continuous line or curve. For example, as is shown in Appendix 1 to this chapter, the so-called *lexicographic ordering* satisfies assumptions 1 to 4, but its indifference sets each consist of only one point. We know that from the point of view of solving optimization problems, continuity is a very important property, and since we shall in effect be using indifference sets (or their geometric representation in the two-good case: indifference curves) to model the consumer’s problem, it is a property we should like them to possess. Hence we make the assumption of continuity.

Assumption 5. Continuity. The graph of an indifference set is a continuous surface.

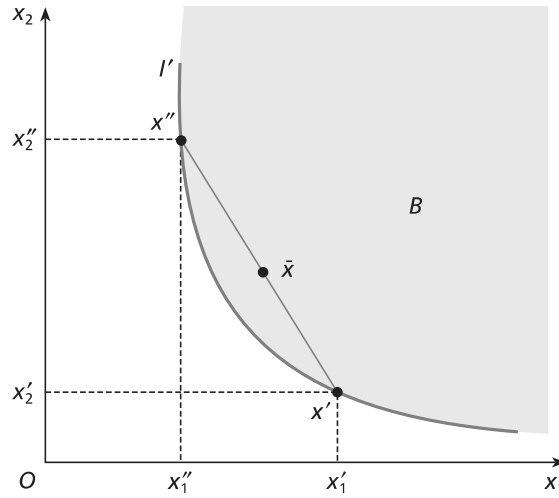
This implies that the surface, or curve in two dimensions, has no gaps or breaks at any point. In terms of the consumer’s choice behaviour, given two goods in his consumption bundle, we can reduce the amount he has of one good, and however small this reduction is, we can always find an increase in the other good which will exactly compensate him, i.e. leave him with a consumption bundle indifferent to the first. The reader should confirm diagrammatically that this is possible only if the indifference surface is everywhere continuous. (See Appendix 2 for a more formal treatment of this assumption.)

We now want to place some restrictions on the shape of the indifference surfaces or curves. From assumption 4 we already know that they must be negatively sloped, and now we say something about their curvature. Recall the earlier definition of the better set of a point x' , as the set of bundles which are preferred or indifferent to x' . Then we make the assumption:

Assumption 6. Strict convexity. Given any consumption bundle x' , its better set is strictly convex.

Figure 2.2 illustrates for the two-good case. The better set for the point x' is the set of points on the indifference curve I' and in the shaded area, and this is drawn as strictly convex. There is an important technical reason for making this assumption: we know (from Appendix E at the end of the book) that, given also that the feasible set is convex, the consumer’s optimal point will as a result be a unique local – and therefore a global – optimum, and this is useful when we analyse the consumer’s responses to changes in the feasible set.

Figure 2.2



There is also a basis for the assumption in terms of economic behaviour. It can be expressed in two ways. From Fig. 2.2 it is clear that, if we move the consumer along the indifference curve leftward from point x' , reducing the quantity of x_1 by small, equal amounts, we have to compensate, to keep him on the indifference curve, by giving him larger and larger increments of x_2 . In other words, the curvature implies that the smaller the amount of x_1 and larger the amount of x_2 held by the consumer, the more valuable are marginal changes in x_1 relative to marginal changes in x_2 . This is a plausible feature of consumer preferences.

A second way of rationalizing the curvature is as follows. In Fig. 2.2, $x' \sim x''$. Consider the straight line joining these two points. Any point on this line, for example $\bar{x}$, is a convex combination of x' and x'' , in that it can be expressed as

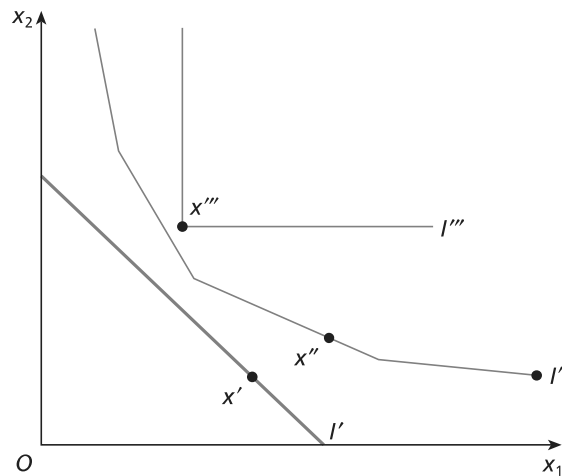
$$\bar{x} = kx' + (1 - k)x'' \quad 1 \geq k \geq 0 \quad [\text{A.1}]$$

i.e. the bundle $\bar{x}$ contains an amount of x_1 given by $kx'_1 + (1 - k)x''_1$, and an amount of x_2 given by $kx'_2 + (1 - k)x''_2$. So, for example, if $k = \frac{1}{2}$, $\bar{x}$ lies halfway along the line, and contains half of x'_1 plus half of x''_1 , and half of x'_2 plus half of x''_2 . We call such a convex combination a mixture of x' and x'' .

It follows from the strict convexity assumption that any mixture along the line will be preferred to x' and x'' (in fact this is the formal *definition* of strict convexity of the better set – see Appendix B). Thus, the consumer always prefers a mixture of two consumption bundles which are indifferent to each other, to either one of those bundles. Again it is argued that this preference for mixtures is a commonly observed aspect of consumer behaviour.

A weaker convexity assumption than assumption 6 can be made: we could assume that the better set is convex but not strictly convex. This means that we allow the possibility of linear segments in the indifference curves, as Fig. 2.3 illustrates. The better sets for points x' , x'' and x''' respectively are each convex but none is strictly convex. Linearity in the indifference curve over some range implies that, within this range, the valuation of marginal decreases in one good relative to marginal increments in the other remains constant – successive equal reductions in the amount of

Figure 2.3



one good are compensated by successive equal increases in the amount of the other. Alternatively, a mixture of two indifferent bundles, in the sense just defined, is indifferent to the two, rather than preferred to them. The reason for excluding such linearity by the strict convexity assumption is, as we shall see, to ensure that the solution to the consumer's problem is a unique point and not a set of infinitely many points.

As a result of these six assumptions, we can represent the preference ordering of the consumer by a set of continuous convex-to-the-origin indifference curves or surfaces, such that each consumption bundle lies on one and only one of them. Moreover, as a result of assumption 4 we can say that bundles on a higher indifference surface are preferred to those on a lower. Thus, the best consumption bundle open to a consumer is the one lying on the highest possible indifference surface. We therefore have part of the analytical apparatus we need to solve the consumer's choice problem.

The utility function

Historically the word 'utility' was used in economics to denote the subjective sensations – satisfaction, pleasure, wish-fulfilment, cessation of need, etc. – which are derived from consumption, and the experience of which is the object of consumption. The economists in the late nineteenth century who were concerned with constructing a theory of consumer choice went further than this definition and regarded utility as something which could be measured in the same way as weight can be measured. They thought it possible to speak of the total quantity of utility derived from consuming a given bundle of goods, of subtracting such quantities from each other, and discussing how these differences changed as consumption varied. Thus was developed the 'law of diminishing marginal utility'. However, even then some of these economists were unhappy about this measurability and it came increasingly under attack as the theory developed. The position which is generally accepted now is that the subjective sensations grouped under the name 'utility' are not capable of

being treated as quantities in this sense. An important reason for the adoption of that position was the demonstration that, for the purpose of constructing a theory of consumer choice, not only the measurement of utility, but the very concept itself, is unnecessary. As we have seen, we can base a theory of choice on the concepts of preference and indifference, and *nothing more is needed* for the theory than the set of indifference curves (or surfaces) with their assumed properties.

However, for some methods of analysis it is useful to have a function which provides a numerical representation of the preference ordering. That is, it is useful to have a rule for associating with each consumption bundle a real number which indicates its place in the ranking. The reason is that we can then apply the standard method of constrained maximization of a function to obtain the solution to the consumer's choice problem.

A suitable rule of association or function can be defined in the following way. On the assumptions made about the consumer's preferences we can partition the consumption bundles into indifference sets and can rank these sets. A rule or function $u(x)$ which assigns a real number u to each bundle x is said to *represent* the consumer's preferences if all bundles in the same indifference set have the same number and bundles in preferred indifference sets have higher numbers, i.e.

- (a) $u(x') = u(x'')$ if and only if $x' \sim x''$
- (b) $u(x') > u(x'')$ if and only if $x' > x''$

Any function satisfying these simple requirements is a *utility function* for the consumer.

A utility function is merely a way of attaching numbers to the consumer's indifference sets such that the numbers increase as higher or more preferred sets are reached. It reflects only the *ordering* of the bundles by the consumer and so is an *ordinal* function. Since we only require that the consumer can rank bundles and the utility function is a numerical representation of this ordering, no significance attaches to the size of the difference between numbers attached to different bundles. We are concerned only with the *sign* of the difference, i.e. whether $u(x') \geq u(x'')$ or whether x' is preferred or indifferent to x'' or x'' preferred to x' .

There are an infinite number of ways of attaching numbers to bundles which are consistent with the requirements (a) and (b) above: the utility function is not unique. For example, given four consumption bundles x' , x'' , x''' , x'''' , any one of the columns in the following table is an acceptable numerical representation of the preference ordering $x' \sim x'' > x''' > x''''$.

	$u(x)$	$v(x)$	$w(x)$
x'	3	10 000	500
x''	3	10 000	500
x'''	2	2	499
x''''	1	1.5	1.9

where $v(x)$ and $w(x)$ denote functions which obey the rule in (a)–(b) above, but which differ from $u(x)$. To put this more formally, we could regard the function $v(x)$ as being derived from $u(x)$ by applying, at each x , some *rule of transformation*, such as, for example, 'When $x = x'$ multiply $u(x')$ by $333\frac{1}{3}$ to obtain $v(x)$ '. That is in general we write:

$$v(x) = T[u(x)] \quad [\text{A.2}]$$

where $T[\cdot]$ denotes the rule of transformation we devise. The only restriction we place on this transformation rule is that when u increases, v must increase, because then v will correctly represent the preference ordering. Such a transformation is called '*positive monotonic*', because v must always increase with u . Hence, we say that the function $u(x)$ is *unique up to a positive monotonic transformation*, meaning that we can always derive another permissible representation of the preference ordering by applying some positive monotonic transformation T to $u(x)$. Examples of such transformations are:

$$\begin{aligned} v(x) &= e^{u(x)} \\ v(x) &= 3 + 2u(x) \\ v(x) &= 5 + \log u(x) \end{aligned} \quad [\text{A.3}]$$

where the transformation T is defined by a simple function. As the table above showed, we do not *have* to define T in such a simple way.

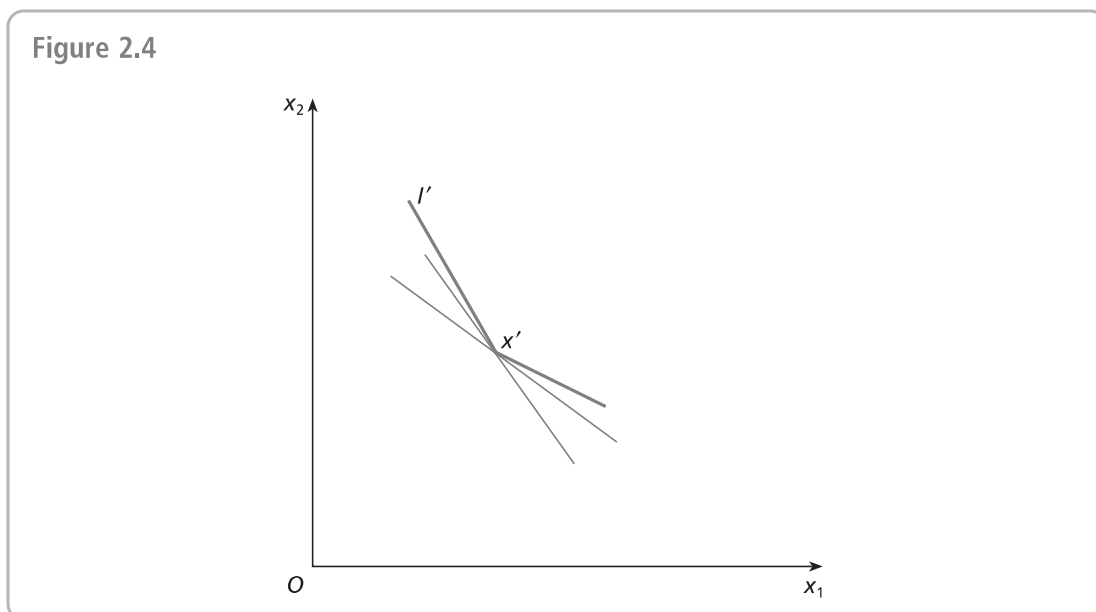
So far we have taken it for granted that a function $u(x)$ which gives a numerical representation of a preference ordering actually does exist. What do we have to assume in order to ensure that the function exists? Consider first assumptions 1–3 above, on completeness, transitivity and reflexivity. Recall that they resulted in a family of indifference sets such that every consumption bundle belonged to one and only one set. We might then reason intuitively that, since that $u(x)$ function effectively assigns numbers to indifference sets, there can be no problem. We would, however, be wrong. It can be shown that we may have a preference ordering satisfying assumptions 1–3 (and 4), but for which no numerical representation exists – we cannot apply to it the rule for assigning numbers to consumption bundles that we set out earlier. An ordering for which this is true is the lexicographic ordering discussed in Appendix 1. The existence of this counter-example tells us that assumptions 1–4 are not sufficient to guarantee existence of a numerical representation of a preference ordering. The further assumption which solves the problem is that of continuity. It can be shown (see Appendix 2) that if assumption 5 holds, so that the indifference surfaces are continuous, a continuous numerical representation $u(x)$ can always be constructed for the preference ordering.

We can now consider the relation between the function $u(x)$ and the indifference sets, which are the fundamental expressions of the consumer's preference ordering. Consider the set of consumption bundles which satisfy

$$u(x) = u^0 \quad [\text{A.4}]$$

where u^0 is some given number. Since these consumption bundles yield the same value of the function they must constitute an indifference set. A set of values of the independent variables in a function which yield a constant value of the function is said to define a *contour* of that function. Hence the indifference sets are contours of the function $u(x)$, and the assumptions 4 and 6 which define the shape of the indifference sets can just as well be interpreted as defining the properties of the contours of $u(x)$. This implies that $u(x)$ is what we called in Appendix B a *strictly quasi-concave* function. In addition, we know that a consumption bundle which yields a higher value of the function than another will always be preferred, and so we can interpret the desire to choose the preferred alternative in some given set of alternatives as equivalent to maximizing the function $u(x)$ over that set. Thus we can represent the consumer's choice problem as one of constrained maximization of a strictly quasi-concave function.

Figure 2.4



In formulating the consumer's choice problem in this way, it is useful if we can use methods of differentiation to find solutions. The assumptions made so far do not imply differentiability: for example, Fig. 2.4 shows a contour which satisfies all the assumptions but is not differentiable at x' – the slope of the contour is not uniquely defined at that point, which is a corner. To rule out such cases, we make the assumption of differentiability (since differentiability implies continuity, we could regard assumption 7 as replacing assumption 5).

Assumption 7. Differentiability. Utility functions are differentiable to any required order.

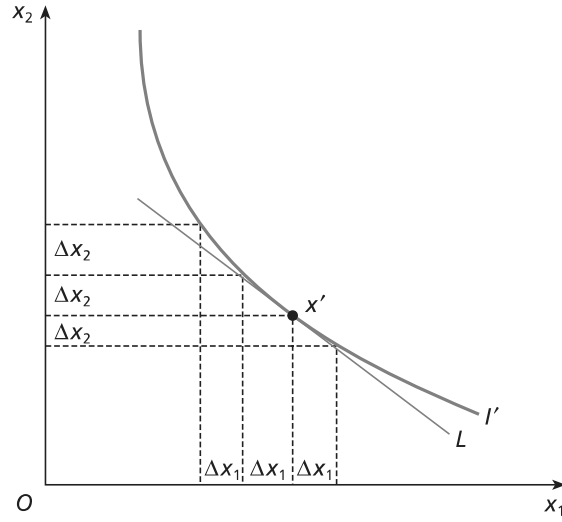
This assumption rules out cases in which the slope of an indifference surface or curve makes a sudden jump, as in Fig. 2.4. We now examine more closely the interpretation of the slope of an indifference curve.

Recall that in discussing assumption 6 we used the idea of successive small reductions in x_1 being compensated by small increments in x_2 just enough to stay on the indifference curve. This can be thought of as defining a 'required rate of compensation', whose (absolute) value increases as we move leftward along the indifference curve. As usual with ratios of *finite* changes, there is an ambiguity arising out of the arbitrariness of the size of the change, and so we find it useful to go to the limit and define the derivative

$$\left. \frac{dx_2}{dx_1} \right|_{u \text{ constant}} = \lim_{\Delta x_1 \rightarrow 0} \left(\frac{\Delta x_2}{\Delta x_1} \right) \quad [\text{A.5}]$$

where the notation on the left-hand side is intended to emphasize that we are constraining the changes in x_1 and x_2 to keep the value of the function u constant. In effect, we view the indifference curve as defining x_2 as a function of x_1 , which could be called an 'indifference function' or 'contour function'. Then the derivative we have defined above is the slope of this function at a point. Figure 2.5 illustrates this.

Figure 2.5



The slope of the tangent L to the indifference curve at x' gives the value of the above derivative at x' . As we take points leftward along the indifference curve, the absolute value of the derivative increases. The figure also shows a sequence of finite changes; the ratio $\Delta x_2/\Delta x_1$ gives the *average* rate of change of x_2 with respect to x_1 over an arc of the curve, and its value will depend on the size of the change Δx_1 .

Important derivatives in economics are always called the marginal something or other, and this is no exception. We define the *marginal rate of substitution* of good 2 for good 1, written MRS_{21} , as

$$MRS_{21} = -\frac{dx_2}{dx_1} \Big|_{u \text{ constant}} \quad [\text{A.6}]$$

The negative sign occurs because we wish MRS_{21} to be positive. Assumption 6 implies that MRS_{21} varies inversely with x_1 . We also define the marginal rate of substitution of good 1 for good 2, written MRS_{12} , as:

$$MRS_{12} = -\frac{dx_1}{dx_2} \Big|_{u \text{ constant}} \quad [\text{A.7}]$$

which refers to the slope of an indifference curve relative to the x_2 axis. Since the two are reciprocals of each other, it is enough to work always with just one of them, and MRS_{21} will usually be taken.

Along an indifference surface we have:

$$u(x) = u^0$$

where u^0 is a given constant. Let u_i , $i = 1, 2, \dots, n$ be the partial derivative $\partial u/\partial x_i$ or the *marginal utility* of good i : the rate at which the utility changes as good i is increased with other goods held constant. If $u_i \neq 0$, the Implicit Function Theorem implies that there is a function $\phi^i(\cdot)$ such that

$$x_i = \phi^i(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) \quad [\text{A.8}]$$

where

$$\frac{\partial \phi^i}{\partial x_j} = - \frac{dx_i}{dx_j} \bigg|_{u \text{ constant}} = \frac{u_j}{u_i} = MRS_{ij} \quad [\text{A.9}]$$

Thus the marginal rate of substitution at a point can be expressed as the ratio of marginal utilities at that point. Since u_i and u_j are in general functions of all n goods, so is MRS_{ij} .

Useful though it is to have this relationship between marginal rates of substitution and partial derivatives of $u(x)$, it is the former which are fundamental. The preference ordering of the consumer uniquely determines the indifference sets and hence the marginal rates of substitution. The partial derivatives, on the other hand, depend on the particular function used to represent the consumer's preferences, or to label the indifference sets.

Properties of marginal utility

If x_i increases with the amounts of all other goods held constant the consumer achieves a better bundle and hence the utility number must increase, so that marginal utility of the i th good is positive: $u_i(x) > 0$. The *sign* of the marginal utility of a good is the same for all numerical representations of the consumer's preferences (i.e. for all utility functions) but the size of the marginal utility is not. If u is a utility function and $v = T[u(x)]$ is a transformation of u with the property that $T' = dT/du > 0$ then $v(x)$ is also a utility function. The partial derivative of v with respect to x_i is

$$\frac{\partial v}{\partial x_i} = v_i = T' \cdot \frac{\partial u}{\partial x_i} = T' u_i \quad [\text{A.10}]$$

and, since by assumption $T' > 0$, the sign of v_i is the sign of u_i but $v_i \neq u_i$ unless $T' = 1$.

The rate of change of marginal utility of x_i with respect to x_i is the second partial derivative of u with respect to x_i : $u_{ii} = \partial^2 u / \partial x_i^2$. Neither the sign nor the magnitude of the rate of change of u_i are the same for all representations of preferences. For example, with the function v considered in the previous paragraph,

$$v_{ii} = \frac{\partial^2 v}{\partial x_i^2} = \frac{\partial}{\partial x_i} (T' u_i) = T' u_{ii} + T'' u_i u_i$$

The sign of v_{ii} is the same as the sign of u_{ii} only if $T'' = d^2 T / du^2 = 0$, but the only restriction on T is that $T' > 0$. Statements about increasing or diminishing marginal utility are therefore meaningless, because we can always find a function to represent the consumer's preferences which contradicts the statement.

Equation [A.9] makes the important point that ratios of marginal utilities are invariant to permissible transformations of the utility function since they must all equal the marginal rate of substitution, which is determined by the consumer's preferences. Using the utility functions u and v above and [A.10], we see that

$$MRS_{ij} = \frac{v_j}{v_i} = \frac{T' u_j}{T' u_i} = \frac{u_j}{u_i} \quad [\text{A.11}]$$

Our warnings about the meaninglessness of statements about the size of changes in utility are valid for the preferences which satisfy the assumptions of this chapter but, as we will see in Chapter 17, if certain additional restricting assumptions about an individual's preferences are made, it becomes sensible to talk of the rate of change of marginal utility. These extra assumptions are unnecessary for our present purposes and so we do not adopt them until they are needed when we study decision-making under uncertainty.

EXERCISE 2A

1. Show that if indifference curves intersect the consumer is inconsistent.
2. Construct a set of indifference curves which satisfy all the assumptions of this section, *except*:
 - (a) one of the 'goods' is in fact a bad; or
 - (b) the consumer may reach a point at which he is satiated with one good but not the other; or
 - (c) the consumer may reach a point at which he is satiated with both goods (a 'bliss point'); or
 - (d) there is a quantity for each good up to which it is a good, and beyond which it is a bad.
 Give concrete examples of goods which may fit each case.
3. Discuss the relationship between the non-satiation assumption and the idea of scarcity which underlies microeconomics.
4. Draw indifference curves relating to:
 - (a) red and blue matches with identical incendiary properties;
 - (b) left and right shoes of the same size, quality, design, etc.;
 and state whether the corresponding utility function is strictly quasi-concave. Comment on the way in which the MRS_{21} varies along these indifference curves. Explain why goods in case (a) are called 'perfect substitutes', and those in (b) 'perfect complements'.
5. Ms A's indifference curves for water and diamonds satisfy the assumption of strict convexity, and she is endowed with a great deal of water and very few diamonds. Which of the following does this imply?
 - (a) Diamonds are more valuable to her than water.
 - (b) She would give up a lot of water to get one more diamond.
 - (c) She would give up more water for an extra diamond than would be the case if she had a combination, indifferent to the first, of less water and more diamonds.

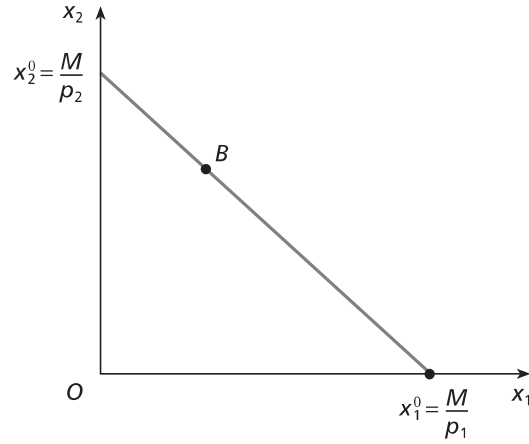
B. The feasible set

We initially assume that the consumer has a given money income M , faces constant prices for all of the goods in the utility function and cannot consume negative quantities of any good. Then (see Appendix A) the consumer's feasible set defined by these assumptions is the set of bundles satisfying

$$\begin{aligned} p_1x_1 + p_2x_2 + \dots + p_nx_n &= \sum p_ix_i \leq M \\ x_1 \geq 0, x_2 \geq 0, \dots, x_n \geq 0 \end{aligned} \quad [\text{B.1}]$$

where p_i is the price of good i .

Figure 2.6



The feasible set in the two-good case is shown in Fig. 2.6 as the triangular area $Ox_1^0x_2^0$. $x_1^0 = M/p_1$ is the maximum amount of x_1 that can be bought with income M at a price of p_1 . x_2^0 is analogously defined. The budget constraint is $p_1x_1 + p_2x_2 \leq M$ in this two-good case, or:

$$x_2 \leq (M - p_1x_1)/p_2 \quad [\text{B.2}]$$

which is satisfied by all points on or below the line B from x_1^0 to x_2^0 . B , the upper boundary of the feasible set, is known as the consumer's *budget line* and is defined by

$$x_2 = (M - p_1x_1)/p_2 \quad [\text{B.3}]$$

The slope of the budget line is

$$\left. \frac{dx_2}{dx_1} \right|_{M \text{ constant}} = -\frac{p_1}{p_2} \quad [\text{B.4}]$$

where the notation on the left-hand side is to remind us that this is the rate at which a consumer with fixed income can exchange x_1 for x_2 on the market. A one-unit reduction in purchases of x_1 reduces expenditure by p_1 , and so, since 1 unit of x_2 costs p_2 , the consumer can buy p_1/p_2 extra units of x_2 . Therefore 1 unit of x_1 exchanges for p_1/p_2 units of x_2 .

The consumer's feasible set has a number of properties relevant for the analysis of the optimal consumption decision (see Appendix B). It is:

- (a) *bounded*, from below by the non-negativity constraints on the x_i and from above by the budget constraint, provided that M is finite and no price is zero. If, for example, $p_1 = 0$ then the budget line would be a line parallel to the x_1 axis through the point $x_2^0 = M/p_2$, and the feasible set would be unbounded to the right: since x_1 would be a free good the consumer could consume as much of it as he wished;
- (b) *closed*, since any bundle on the budget line B or the quantity axes is available;

- (c) *convex*, since for any two bundles x' and x'' in the feasible set, any bundle $\bar{x}$ lying on a straight line between them will also be in the feasible set. Since $\bar{x}$ lies between x' and x'' , and they both satisfy the non-negativity constraints, $\bar{x}$ will also satisfy these constraints. $\bar{x}$ will cost no more than the consumer's income: lying between x' and x'' it must cost no more than the more expensive of them, say x' . But since x' lies within the feasible set, so must $\bar{x}$. Hence $\bar{x}$ is in the feasible set;
- (d) *non-empty*: provided that $M > 0$ and at least one price is finite the consumer can buy a positive amount of at least one good.

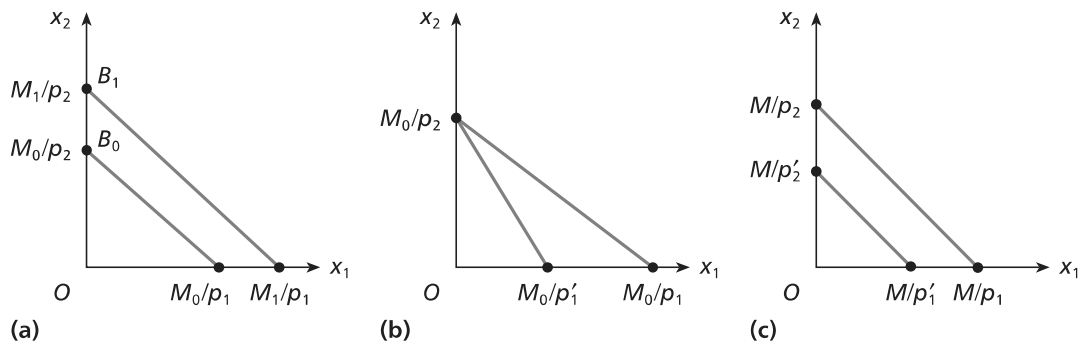
Consider the effects of changes in M and p_i on the feasible set, in preparation for section D where we examine their effects on the consumer's optimal choice. If money income increases from M_0 to M_1 , the consumer's feasible set expands as the budget line moves outward parallel with its initial position, as in Fig. 2.7(a). With $M = M_0$ the intercepts of the budget line B_0 on the x_1 and x_2 axes respectively are M_0/p_1 and M_0/p_2 and with $M = M_1$ they are M_1/p_1 and M_1/p_2 . A doubling of M , for example, will double the value of the intercepts, since $M_1/p_2 = 2M_0/p_2$ when $M_1 = 2M_0$. The slope of the budget line is $-p_1/p_2$ and is unaffected by changes in M .

Consider next an increase in p_1 , as shown in Fig. 2.7(b). Since M and p_2 are unchanged the budget line will still have the same M/p_2 intercept on the x_2 axis. An increase in p_1 will cause the budget line to pivot about M_0/p_2 and become more steeply sloped as p_1/p_2 becomes larger. In Fig. 2.7(b) a rise in p_1 to p'_1 shifts the x_1 intercept from M_0/p_1 to M_0/p'_1 where $M_0/p_1 > M_0/p'_1$ since $p_1 < p'_1$.

Equal proportionate increases in all prices cause the budget line to shift inwards towards the origin as in Fig. 2.7(c). Suppose p_1 and p_2 increase from p_1 and p_2 to kp_1 and kp_2 where $k > 1$. Then the slope of the new budget line is unchanged: $-kp_1/kp_2 = -p_1/p_2$, and the new intercepts are $M/kp_1 < M/p_1$ and $M/kp_2 < M/p_2$.

Finally, if all prices and M change in the same proportion the budget line is unchanged. The intercept on the i th axis after all prices and M change by the factor k is $kM/kp_i = M/p_i$ so the intercept is unaffected, as is the slope, which is $-kp_1/kp_2 = -p_1/p_2$.

Figure 2.7



EXERCISE 2B

1. Suppose that the price of one of the commodities bought by the consumer rises as he buys larger quantities. What effect will this have on his feasible set? What interpretation can be given to the slope of his budget line? Can you show in the diagram the relationship between the average price (expenditure divided by quantity bought) and the marginal price?
2. Many public utilities sell their products on multi-part tariffs. The consumer must pay a connection charge for the right to consume (say) electricity, irrespective of the amount consumed. The price paid for the first n units will exceed the price paid for any units consumed in excess of n . Draw the feasible set for the consumer, with electricity on one axis and a consumption good on the other. Distinguish between the average and marginal prices of electricity and investigate the effects of changes in the connection charge and the price of electricity.
3. Draw the feasible set of the consumer in Question 2a, Exercise 2A, assuming that the 'bad' is garbage and that there is a given price per bag of garbage removed, and a given amount of garbage produced per period by the consumer.

C. The consumption decision

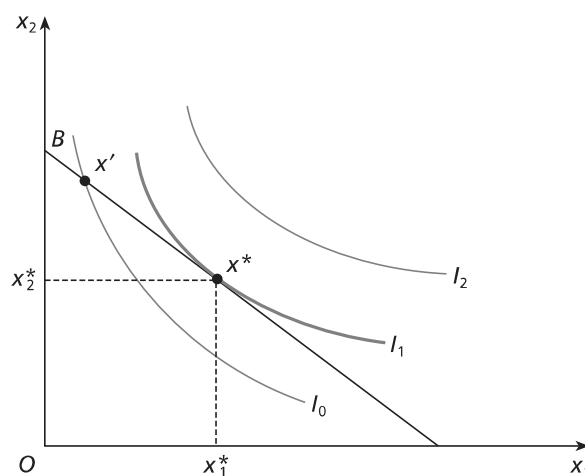
Given the assumptions of the previous two sections, the consumer's problem of choosing the most preferred bundle from those available can be formally stated as

$$\max_{x_1, \dots, x_n} u(x_1, x_2, \dots, x_n) \quad \text{s.t.} \quad \sum_i p_i x_i \leq M, \quad x_i \geq 0, \quad i = 1, \dots, n \quad [\text{C.1}]$$

We can derive the conditions which the solution to this problem must satisfy by a diagrammatic analysis of the two-good case. We leave to the latter part of this section a brief confirmation of our results using more rigorous methods.

From the assumptions of section A we can represent the consumer's preferences by a utility function which has indifference curves or contours like those of Fig. 2.8.

Figure 2.8



All commodities are assumed to have positive marginal utility so that bundles on higher indifference curves are preferred to those on lower indifference curves. This assumption (a consequence of assumption 4 in section A) also means that the consumer will spend all his income since he cannot be maximizing u if he can buy more of some good with positive marginal utility. The consumer will therefore choose a bundle on his budget line B .

In Fig. 2.8 there is a *tangency solution* where the optimal bundle x^* is such that the highest attainable indifference curve I_1 is tangent to the budget line and the consumer consumes some of both goods. The slope of the indifference curve is equal to the slope of the budget line at the optimum:

$$\left. \frac{dx_2}{dx_1} \right|_{u \text{ constant}} = \left. \frac{dx_2}{dx_1} \right|_{M \text{ constant}}$$

The negative of the slope of the indifference curve is the marginal rate of substitution MRS_{21} ; and the negative of the slope of the budget line is the ratio of the prices of x_1 and x_2 . Hence the consumer's equilibrium condition can be written as

$$MRS_{21} = \frac{u_1}{u_2} = \frac{p_1}{p_2} \quad [\text{C.2}]$$

The consumer is in equilibrium (choosing an optimal bundle) when the rate at which he *can* substitute one good for another on the market is equal to the rate at which he is just content to substitute one good for another.

We can interpret this property of the optimal choice in a somewhat different way. If the consumer spent an extra unit of money on x_1 he would be able to buy $1/p_1$ units of x_1 . $u_1 \Delta x_1$ is the gain in utility from an additional Δx_1 units of x_1 . Hence u_1/p_1 is the gain in utility from spending an additional unit of money on x_1 . u_2/p_2 has an analogous interpretation. The consumer will therefore be maximizing utility when he allocates his income between x_1 and x_2 so that the marginal utility of expenditure on x_1 is equal to the marginal utility of expenditure on x_2 :

$$\frac{u_1}{p_1} = \frac{u_2}{p_2} \quad [\text{C.3}]$$

This is exactly the condition obtained by multiplying both sides of [C.2] by u_2/p_1 .

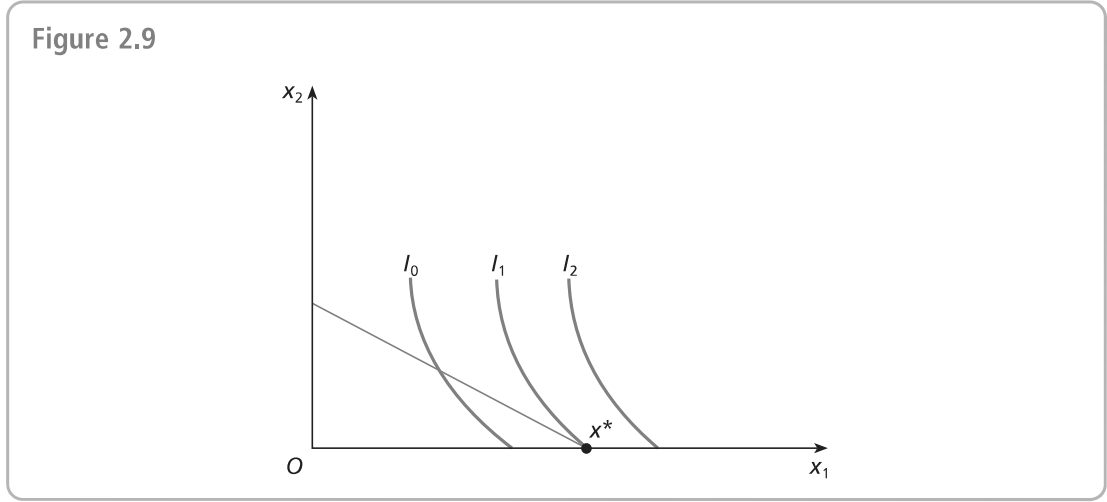
If the consumer's income were increased by a small amount he would be indifferent between spending it on x_1 or x_2 : in either case utility would rise by $u_1/p_1 = u_2/p_2$. Hence, if we call the rate at which the consumer's utility increases as income increases the marginal utility of income, denoted by u_M , we have

$$\frac{u_1}{p_1} = \frac{u_2}{p_2} = u_M \quad [\text{C.4}]$$

A more plausible optimum when there are many goods would be a *corner point solution*, where the optimal bundle x^* does not contain positive amounts of all goods, as in Fig. 2.9 where no x_2 is purchased. In this case the indifference curve at x^* is steeper than the budget line, i.e. has a smaller slope (remembering that the indifference curve and the budget line are negatively sloped). Hence

$$\left. \frac{dx_2}{dx_1} \right|_{u \text{ constant}} < \left. \frac{dx_2}{dx_1} \right|_{M \text{ constant}} \quad [\text{C.5}]$$

Figure 2.9



and therefore

$$\left. \frac{-dx_2}{dx_1} \right|_{u \text{ constant}} = MRS_{12} = \frac{u_1}{u_2} > \frac{p_1}{p_2} = \left. \frac{-dx_2}{dx_1} \right|_{M \text{ constant}} \quad [\text{C.6}]$$

Rearranging, this condition can be written

$$u_M = \frac{u_1}{p_1} > \frac{u_2}{p_2} \quad [\text{C.7}]$$

The marginal utility of expenditure on the good purchased, x_1 , is greater than the marginal utility of expenditure on x_2 , the good not purchased. Because of the higher marginal utility of expenditure on x_1 than on x_2 the consumer would like to move further down the budget line substituting x_1 for x_2 but is restrained by the fact that consumption of negative amounts of x_2 is not possible.

A more formal analysis

Since the consumer's preferences satisfy the assumptions of section A, the objective function in problem [C.1] above is continuous and strictly quasi-concave. From section B the feasible set for the problem, defined by the budget and non-negativity constraints, is non-empty, closed, bounded and convex. From the Existence, Local-Global and Uniqueness Theorems (see Appendices C, D, E), the consumer's optimization problem has a unique solution and there are no non-global local solutions.

Since there is at least one good with positive marginal utility the consumer spends all income and hence the budget constraint can be written as an equality constraint: $M - \sum p_i x_i = 0$. If we assume that the solution will be such that some of all goods will be consumed ($x_i^* > 0$ ($i = 1, \dots, n$) where x_i^* is the optimal level of x_i), then the non-negativity constraints are non-binding and we have a problem to which can be applied the method of Lagrange outlined in Appendix G. The Lagrange function derived from [C.1] is

$$L = u(x_1, \dots, x_n) + \lambda[M - \sum p_i x_i] \quad [\text{C.8}]$$

and the first-order conditions for a solution to [C.1] are

$$\frac{\partial L}{\partial x_i} = u_i - \lambda p_i = 0 \quad i = 1, \dots, n \quad [\text{C.9}]$$

$$\frac{\partial L}{\partial \lambda} = M - \sum p_i x_i^* = 0 \quad [\text{C.10}]$$

If [C.9] is rewritten as $u_i = \lambda p_i$ and the condition on good i is divided by that on good j , we have

$$\frac{u_i}{u_j} = \frac{p_i}{p_j} \quad [\text{C.11}]$$

or: the marginal rate of substitution between two goods is equal to the ratio of their prices as in condition [C.3] above. Alternatively, [C.9] can be rearranged to give

$$\frac{u_1}{p_1} = \frac{u_2}{p_2} = \dots = \frac{u_n}{p_n} = \lambda \quad [\text{C.12}]$$

which is the n -good extension of the condition [C.4] derived earlier.

The value of the Lagrange multiplier λ is the rate at which the objective function increases as the constraint parameter increases. In this case the objective function is the utility function and the constraint parameter is the individual's money income so that λ is the rate at which utility increases as money income increases:

$$\lambda = \frac{du^*}{dM} = u_M \quad [\text{C.13}]$$

The Lagrange multiplier can be interpreted as the marginal utility of money income. This interpretation is supported by [C.12] since, as we argued above, u_i/p_i is the rate at which utility increases as more money is spent on good i .

Corner solutions

If the assumption that $x_i^* > 0$ for all i is dropped, the first order conditions for [C.1] are derived by maximization of the Lagrangean [C.8] subject to the direct non-negativity constraints on the choice variables. The conditions which must be satisfied by a solution to C.1 are

$$\begin{aligned} \frac{\partial L}{\partial x_i} = u_i - \lambda^* p_i &\leq 0, & x_i^* &\geq 0, & x_i^*(u_i - \lambda^* p_i) &= 0 \\ i &= 1, 2, \dots, n \end{aligned} \quad [\text{C.14}]$$

plus condition [C.10].

If [C.14] is rearranged to give

$$\lambda^* \geq \frac{u_i}{p_i}, \quad x_i^* \geq 0, \quad x_i^* \left(\frac{u_i}{p_i} - \lambda^* \right) = 0 \quad [\text{C.15}]$$

it can be given a straightforward economic interpretation: if the marginal utility of expenditure on good i , (u_i/p_i) , is less than the marginal utility of money at the optimal point, λ^* , then good i will not be bought since the consumer will get greater utility by expenditure on other goods. The same result can be derived from [C.7], where $x_2 = 0$, since $u_1/p_1 = u_M > u_2/p_2$, or $u_2 - u_M p_2 < 0$.

EXERCISE 2C

1. If a consumer buys electricity on a multi-part tariff, as in Question 2, Exercise 2B, are the conditions of the Existence, Local–Global and Uniqueness Theorems satisfied?
2. Derive and interpret the equilibrium conditions for the types of preferences postulated in Questions 2 and 4, Exercise 2A.
(Note: what must be assumed about the price of a ‘bad’?)
3. Explain why a consumer would:
 - (a) not choose a point inside his budget line;
 - (b) not choose the bundle x' in Fig. 2.8.
4. Suppose that, as well as paying a price per unit of a good, the consumer has to pay a ‘transactions cost’ for using a market. Analyse the implications for the consumer’s optimal choice of assuming:
 - (a) the transactions cost is paid as a lump sum;
 - (b) the transactions cost is proportional to price but independent of the quantity bought;
 - (c) the transactions cost is charged per unit of the good bought, but decreases the greater the amount bought.
5. Is the marginal utility of money income, λ^* or u_M , uniquely defined?

D. The comparative statics of consumer behaviour

The solution to the consumer’s optimization problem depends on the consumer’s preferences, prices and money income. We can write the solution, which we call the *demand for goods*, as a function of prices and money income:

$$x_i^* = D_i(p_1, p_2, \dots, p_n, M) = D_i(p, M) \quad i = 1, \dots, n \quad [\text{D.1}]$$

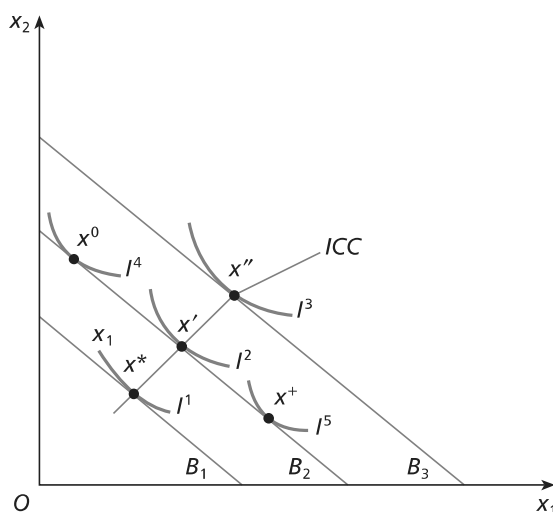
where $p = (p_1, p_2, \dots, p_n)$ is the vector of prices, and the form of the *Marshallian demand function* D_i depends on the consumer’s preferences.

The properties of feasible sets and the objective function enable us to place restrictions on the form of the demand functions. First, provided that p, M are finite and positive, the optimization problem must have a solution, since the requirements of the Existence Theorem are satisfied. Second, the differentiability of the indifference curves and the linearity of the budget constraint imply that the optimal bundle will vary continuously in response to changes in prices and income, and that the demand functions are *differentiable*. Third, the conditions of the Uniqueness Theorem are satisfied and so the demand relationships are functions rather than correspondences: a *unique* bundle is chosen at each (p, M) combination.

We now consider the *comparative statics* properties of the model. We investigate the effects of changes in the exogenous variables (prices, money income) on the equilibrium values of the endogenous variables (the consumer’s demand for goods). We want to predict what happens to the optimal bundle $x^* = (x_1^*, x_2^*, \dots, x_n^*) = (D_1, D_2, \dots, D_n)$ as the feasible set varies.

We consider first changes in the consumer’s money income. In Fig. 2.10, B_1 is the initial budget line, x^* the initial bundle chosen. An increase in M , with p_1, p_2 constant, will shift the budget line outward parallel with itself, say to B_2 where x' is chosen. A further increase in M will shift the budget line to B_3 where x'' is chosen.

Figure 2.10

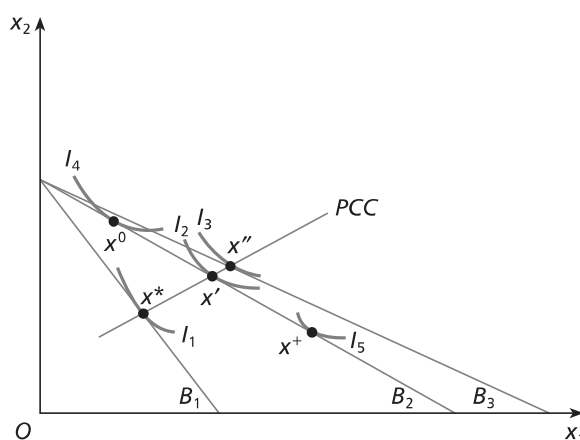


The *income consumption curve* is the set of optimal points traced out as income varies in this way, with prices constant. In the case illustrated both x_1 and x_2 are *normal goods*, for which demand increases as money income rises. However, with different preferences the consumer might have chosen x^0 or x^+ on B_2 . If x^0 had been chosen (if I^4 and not I^2 had been the consumer's indifference curve) then the demand for x_1 would have fallen as money income rose. x_1 would then be known as an *inferior good*. A rise in M may lead to a rise, a fall, or no change in the demand for a good. Without knowledge of preferences we cannot predict whether a particular good will be inferior or normal. The theory of consumer behaviour cannot be tested by considering the effect of changes in M on the demand for a single good, since any effect is compatible with the theory.

The theory does predict, however, that *all* goods cannot be inferior. If the consumer reduces demand for all goods when income rises he will be behaving inconsistently. To show this, let x^* be the bundle chosen with an initial money income of M_1 and x' the bundle chosen when money income rises to M_2 . If $x' \ll x^*$ i.e. if the demand for all goods is reduced, then x' must cost less than x^* since prices are held constant. x' was therefore available when x^* was chosen. But when x' was chosen x^* was still attainable (since money income had increased). The consumer therefore preferred x^* over x' with a money income of M_1 and x' over x^* with money income $M_2 > M_1$. He is therefore inconsistent: his behaviour violates the *transitivity assumption* of section A, and our model would have to be rejected.

We now turn to the effects of changes in prices on the consumer's demands. Figure 2.11 shows the implications of a fall in the price of x_1 with money income held constant. B_1 is the initial budget line, x^* the initial optimal bundle. A fall in p_1 , say from p_1 to p'_1 , causes the budget line to shift to B_2 . x' is the optimal bundle on B_2 , x'' the optimal bundle on B_3 , which results from a further fall in p_1 from p'_1 to p''_1 . The *price consumption curve* (PCC) is traced out as the set of optimal bundles as p_1 varies. In this case the demand for both goods increases as p_1 falls. However, with different preferences the optimal bundle might have been x^0 or x^+ on B_2 . If x^0 was the optimal bundle with $p_1 = p'_1$ then x_1 would be a *Giffen good*, the demand for which falls as its

Figure 2.11



price falls. We conclude that the demand for a good may fall, rise or remain unchanged as a result of a change in a price facing the consumer. Once again the model yields no definite (refutable) prediction about the effect on a *single* endogenous variable (the demand for a good) of a change in *one* of the exogenous variables (in this case a price). It is again possible, however, to predict (by reasoning similar to that employed in the case of a change in M) that a fall in price will not lead to a reduction in demand for *all* goods, and the reader should supply the argument.

Income and substitution effects

The analysis of the effect of price changes on the consumer's demands (optimal choices) has suggested that demand for a good may increase, decrease or remain unchanged, when its price rises; in other words anything may happen. We now examine the effect of a change in the price of good 1 in more detail in order to see if it is possible to make more definite (refutable) predictions. We proceed by making a conceptual experiment. All we can actually observe is the change in quantity demanded following a price change. However, we can carry out a hypothetical analysis which decomposes the overall demand change into two components. We then use this decomposition to say something more definite about consumer behaviour.

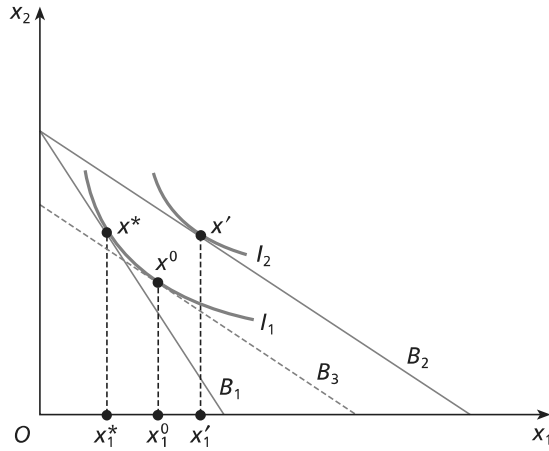
In Fig. 2.12, it can be seen that the fall in price of good 1 does two things:

- it reduces the expenditure required to achieve the initial utility level I_1 , allowing the higher utility level I_2 to be achieved with the same expenditure. There has been an increase in the consumer's *real income*;
- it changes the relative prices facing the consumer.

In Fig. 2.12 we accordingly break down the change in demand for x_1 into:

- the income effect*, which is the change resulting solely from the change in real income, with relative prices held constant; and
- the own substitution effect*, which results solely from the change in p_1 with real income held constant.

Figure 2.12



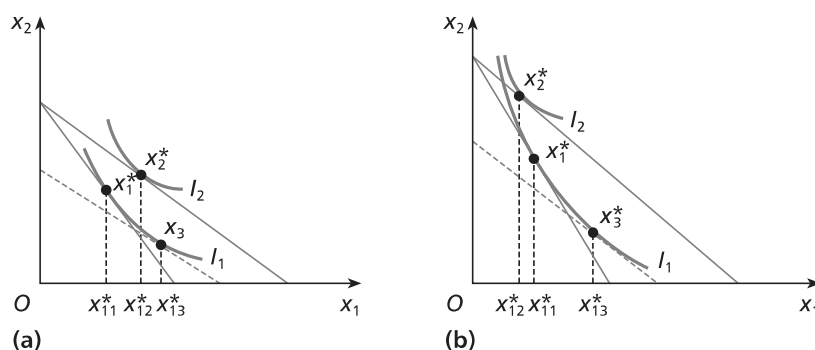
x^* and x' are the optimal bundles before and after the fall in p_1 , B_1 and B_2 the corresponding budget lines. The *compensating variation* in money income is that change in M which will make the consumer just as well off after the price fall as he was before. In other words, there will be some reduction in M after the price fall which will 'cancel out' the real income gain and return the consumer to the initial indifference curve I_1 . The budget line is shifted inwards (reducing M) parallel with the post-price fall budget line B_2 until at B_3 it is just tangent to the original indifference curve I_1 . The consumer confronted with this budget line would choose bundle x^0 . The difference between x^* and x^0 is due to a change in relative prices with real income (utility) held constant. The difference between x^0 and x' is due to the change in money income with relative prices held constant. x_1^* , x_1' and x_1^0 are the amounts of x_1 contained in the bundles x^* , x' , x^0 and

- (a) $x_1^0 - x_1^*$ is the own substitution effect;
- (b) $x_1' - x_1^0$ is the income effect;
- (c) $(x_1^0 - x_1^*) + (x_1' - x_1^0) = x_1' - x_1^*$ is the total *price effect*.

The purpose of carrying out this experiment in hypothetical compensation is to demonstrate that the own substitution effect will always be positive in the case of a price fall and negative for a price rise. The absolute value of the slope of the indifference curve declines from left to right, i.e. as more x_1 and less x_2 is consumed the curve flattens. The fall in p_1 flattens the slope of the budget line, and hence the budget line B_3 *must* be tangent with I_1 to the right of x^* , i.e. at a bundle containing more x_1 .

The income effect is positive in the particular case illustrated in Fig. 2.12. The income effect reinforces the own substitution effect since x' contains more x_1 than x^0 . If x_1 had been inferior the income effect of the price fall would have been negative and in the opposite direction to the own substitution effect, so that the price effect would be smaller than the own substitution effect. In Fig. 2.13(a) the income effect partially offsets the substitution effect but the price effect is still positive: a fall in p_1 leads to a rise in the demand for x_1 . In Fig. 2.13(b) the negative income effect more than offsets the positive substitution effect and x_1 is a Giffen good. Hence inferiority is a necessary, but not sufficient, condition for a good to be a Giffen good.

Figure 2.13



This decomposition of the price effect has generated two further predictions:

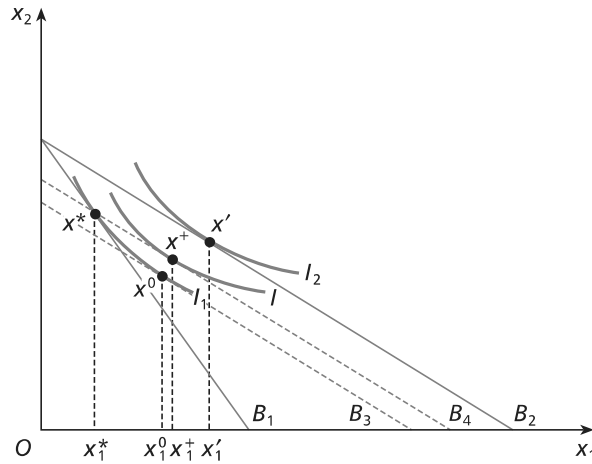
1. A normal good cannot be a Giffen good. Hence, if we observe that a consumer increases demand for a good when money income rises (other things including prices being held constant), we would predict that, if its price should fall, he will want to buy more of it. If we then observe that he reduces his demand for the good when its price falls (and all other prices are constant and his money income is reasonably close to its original level), then the optimizing model of consumer behaviour has yielded a false prediction.
2. The own substitution effect is always of opposite sign to the price change.

The above decomposition of the price effect into an income and substitution effect is based on the definition, made by Hicks, of unchanged *real income* as an unchanged *utility level*. Slutsky suggested an alternative definition of a constant real income as the ability to purchase the bundle of goods bought before the price change. This *constant purchasing power* definition has the advantage that it does not require detailed knowledge of the consumer's indifference map.

Figure 2.14 reproduces Fig. 2.11 with some additions to show the relationship between the Hicks and Slutsky definitions of a constant real income. The budget line B_4 just enables the consumer to buy x^* , the initially optimal bundle, at the lower price of p_1 . Confronted with this budget line, the consumer actually chooses x^+ . The price effect has been decomposed into an income effect ($x_1^+ - x_1^*$) and an own substitution effect ($x_1^+ - x_1^*$). The income effect will again be positive, negative or zero depending on the form of the indifference map. The substitution effect will, as in the Hicksian case, always lead to a rise in demand for a good whose price has fallen. x^+ cannot lie to the left of x^* on B_4 because this would mean that the consumer is now choosing x^+ when x^* is still available, having previously rejected x^+ in favour of x^* . The transitivity assumption would be violated by such behaviour. The Slutsky definition yields a prediction (the sign of the substitution effect) which can be tested without specific knowledge of the consumer's indifference curves to 'cancel out' the income effect.

Our consideration of the comparative static properties of the model has shown that it does not yield refutable predictions about the overall change in demand for individual goods induced by *ceteris paribus* changes in a price or money income. In other words,

Figure 2.14



$$\frac{\partial x_i^*}{\partial p_j} = \frac{\partial D_i}{\partial p_j} \cong 0 \quad i, j = 1, 2, \dots, n$$

and

$$\frac{\partial x_i^*}{\partial M} = \frac{\partial D_i}{\partial M} \cong 0 \quad i = 1, 2, \dots, n$$

for every good and price. Only by considering the effect of changes in p_j or M on *all* goods, or by considering the effect of changes in p_j and M on a single good or by making more specific assumptions about the consumer's preferences can definite predictions be generated.

Consider, however, the consequences of equal proportionate changes in all prices and M . Suppose M increases to kM ($k > 1$) and prices to kp_1 and kp_2 . The slope of the budget line will be unaffected. The intercept on the x_1 axis is M/p_1 before the changes in M and prices and $kM/kp_1 = M/p_1$ after the change. Similarly for the intercept on the x_2 axis. Hence the equal proportionate changes in M and all prices alter neither the slope nor the intercepts on the budget line and so the feasible set is unaltered. If the feasible set is unchanged then so is the optimal bundle.

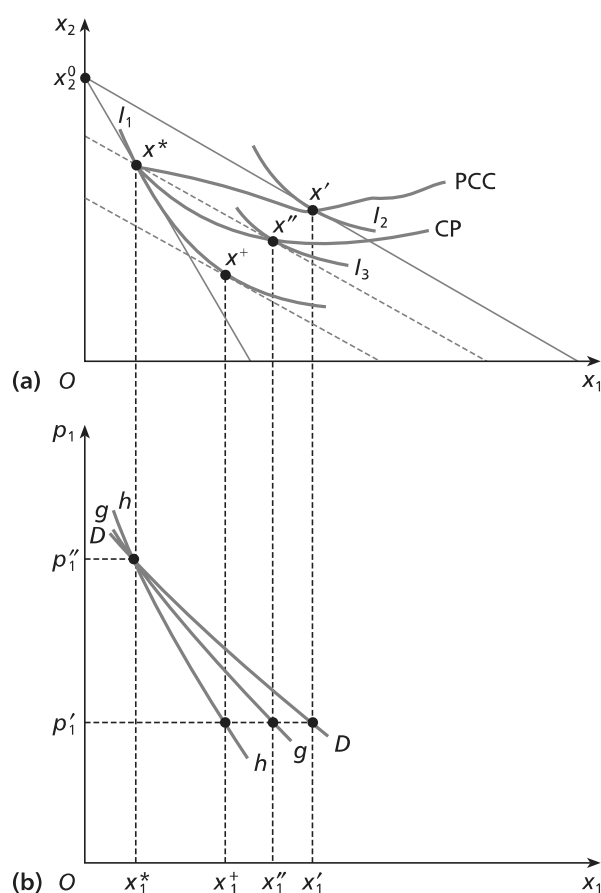
The model therefore predicts that the consumer will not suffer from *money illusion*; he will not alter his behaviour if his purchasing power and relative prices are constant, irrespective of the absolute level of prices and money income. More formally, the demand function D_i for every commodity is *homogeneous of degree zero* in prices and money income, since we have

$$x_i^* = D_i(kp, kM) = k^0 D_i(p, M) = D_i(p, M) \quad [\text{D.2}]$$

Demand curves

We complete this section on comparative statics by deriving the demand curve from the utility maximization model. The individual's *demand curve* for a good shows how desired or planned purchases of it vary as its price varies, other prices and income being held constant. As we have seen, a distinction can be drawn between constant

Figure 2.15



real and constant money income and there are also two possible definitions of constant real income. Figure 2.15 shows the derivation of three demand curves, corresponding to the different assumptions about what is held constant, from the consumer's indifference map. The upper part of Fig. 2.15 is Fig. 2.14 with two additions. The PCC (price consumption curve) shows the bundles chosen as p_1 varies with M constant (i.e. as the budget line pivots through x_2^0). The constant purchasing power consumption curve (CP) shows the bundles chosen as p_1 varies, with the consumer's money income varying so as just to enable the consumer to purchase the original bundle x^* (i.e. the Slutsky definition of constant real income is adopted and so the budget line pivots through x^*). The indifference curve I_1 shows how consumption varies as p_1 varies, with M varying to keep the consumer's utility level constant (i.e. the Hicks definition of constant real income is adopted and so the budget line slides round I_1). These three curves therefore show the change in the demand for x_1 (and x_2) as p_1 changes, with income (variously defined) and p_2 held constant. The lower half of the figure uses the information contained in the three curves to plot demand curves for x_1 .

The *Marshallian constant money income demand curve* DD shows the effect of changes in p_1 with M (and p_2) held constant. It plots the information contained in

the price consumption curve. For example, a fall in p_1 from p_1'' to p_1' with M constant causes the consumer to shift from bundle x^* to x' and his demand for x_1 to rise from x_1^* to x_1' .

The *constant purchasing power demand curve* gg corresponds to the CP curve. The fall in p_1 from p_1'' to p_1' with purchasing power constant causes the consumer to shift from x^* to x'' and his demand to increase from x_1^* to x_1'' .

The *Hicksian constant utility demand curve* hh is derived from the indifference curve I_1 . The fall in p_1 with utility constant at its initial level $u(x^*)$ causes the consumer to shift from x^* to x^+ , and his demand to increase from x_1^* to x_1^+ .

The constant money income demand curve plots the whole price effect and the other two curves plot only the two versions of the substitution effect. Hence the constant utility and purchasing power demand curves will be steeper than the constant money income demand curve when x_1 is a normal good, because they do not plot the income effect of the price change. When x_1 is inferior the relative steepness of the various demand curves is reversed. This analysis is taken further in Chapter 3, with the help of duality theory.

EXERCISE 2D

1. Derive the demand curve of the consumer of Question 3, Exercise 2B, for garbage disposal. Decompose the effects of a price change into income and substitution effects.
2. Examine the responses of an electricity consumer to changes in the connection charge and prices of electricity.
3. Examine the income and substitution effects in the cases given in Question 4, Exercise 2A.
4. Explain the difference between the Hicks and Slutsky definitions of real income, and apply this to explain why, in Fig. 2.15, the demand curve hh is steeper than the demand curve gg .
5. Why do we decompose the price effect into income and substitution effects?
6. Examine the properties of the demand functions of a consumer with the following utility functions:
 - (a) $u(x) = x_1^{\alpha_1} x_2^{\alpha_2} \dots x_n^{\alpha_n}$ ($\alpha_i > 0$, all i ; $\sum \alpha_i = 1$) (Cobb–Douglas)
 - (b) $u(x) = (x_1 - k_1)^{\alpha_1} (x_2 - k_2)^{\alpha_2} \dots (x_n - k_n)^{\alpha_n}$ ($\alpha_i > 0$, all i ; $\sum \alpha_i = 1$; $k_i > 0$, all i) (Stone–Geary)
 - (c) $u(x) = \sum_i f_i(x_i)$ ($f_i' > 0$) (additive separable)
 - (d) $u(x) = f(x_1) + x_2$ ($f' > 0$; $f'' < 0$) (quasi-linear)

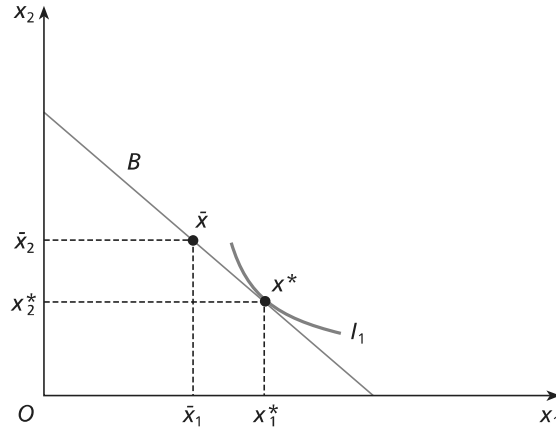
What interpretation can be given to the k_i in case (b)?

E. Offer curves and net demand curves

We now consider the case of a consumer who has preferences satisfying the assumptions of section A, and is endowed, not with a given money income, but with fixed amounts of commodities which she can consume or sell on the market in order to finance purchases of other commodities. The feasible set is defined by the non-negativity requirements on consumption and by the constraint that the market value of the bundle consumed cannot exceed the market value of the consumer's initial endowments. Her budget constraint is therefore:

$$\sum p_i x_i \leq \sum p_i \bar{x}_i = W \quad i = 1, 2, \dots, n \quad [\text{E.1}]$$

Figure 2.16



where $\bar{x}_i$ is her *initial endowment of good i* . $\sum p_i \bar{x}_i = W$ is the market value of the initial endowment, or the proceeds which could be obtained if the consumer sold all her initial endowments at the ruling market prices.

If x_i (the amount of commodity i consumed) differs from her endowment $\bar{x}_i$, then

- (a) if $\hat{x}_i = x_i - \bar{x}_i > 0$, the consumer buys $\hat{x}_i$ of commodity i ;
- (b) if $\hat{x}_i = x_i - \bar{x}_i < 0$, she sells $\hat{x}_i$ of the commodity.

$\hat{x}_i$ is the *net demand* for commodity i . We can rewrite the budget constraint as:

$$\sum p_i \hat{x}_i = \sum p_i (x_i - \bar{x}_i) \leq 0 \quad [\text{E.2}]$$

The sum of her expenditures on the quantities of goods she buys (which will be a positive component of the overall sum) cannot exceed the sum of the proceeds from the quantities of goods she sells (a negative component of the overall sum).

In the two-good case shown in Fig. 2.16 the budget line B is defined by

$$p_1 x_1 + p_2 x_2 = p_1 \bar{x}_1 + p_2 \bar{x}_2 = W \quad \text{or} \quad p_1 (x_1 - \bar{x}_1) + p_2 (x_2 - \bar{x}_2) = 0 \quad [\text{E.3}]$$

and has a slope of $-(p_1/p_2)$. B must pass through $\bar{x} = (\bar{x}_1, \bar{x}_2)$, the endowed bundle, since whatever prices she faces the consumer will always have the possibility of consuming her endowment, i.e. neither buying nor selling on the market. The feasible set is similar in shape to the case of the consumer endowed with a fixed money income M , but there are significant differences as regards the effect of changes in prices:

- (a) The market value of the endowment $W = \sum p_i \bar{x}_i$ will increase or decrease as the price of a commodity increases or decreases.
- (b) Since the consumer is always able to consume her initial endowment vector $\bar{x}$, a change in a single price will cause the budget line to rotate through $\bar{x}$, rather than through an intercept on one of the axes.
- (c) An equal proportionate change in all prices leaves the budget line unaffected, though the nominal value of the endowments varies in the same proportion. The budget line must still pass through $\bar{x}$, and its slope will be unaffected by such

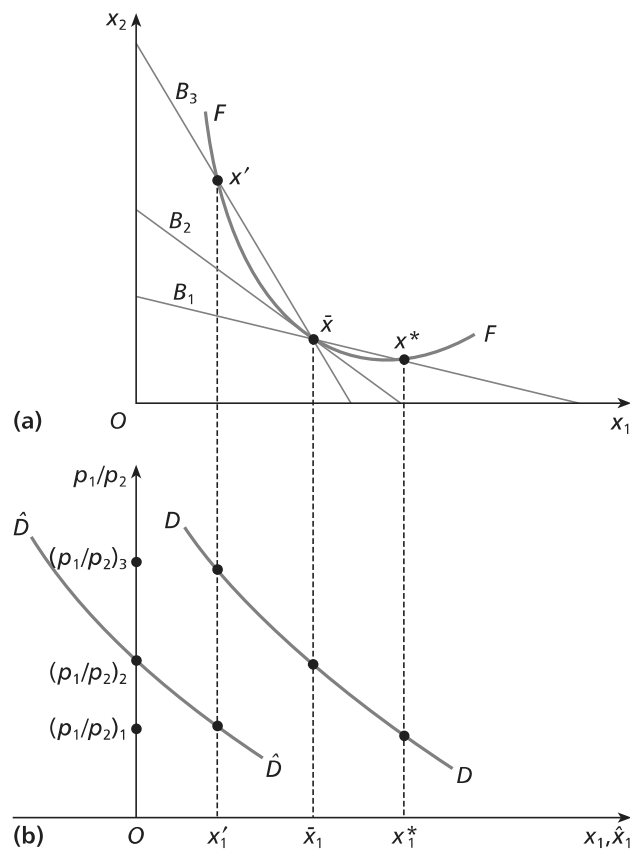
price changes and hence its position is unchanged. Only change in *relative* prices or in the initial endowments will alter the consumer's feasible set and therefore the consumer's demand or supply of a commodity. In the terminology of section D, her demand functions will again be *homogeneous of degree zero in prices*.

It is clear from the way in which the budget constraint [E.1] was written that the consumer's optimization problem in this case is formally identical to that considered previously, so we will not dwell on the equilibrium conditions and the possibility of corner solutions. We restrict ourselves to examining the comparative static properties of tangency solutions and the derivation of supply and demand curves.

In Fig. 2.16, $x^* = (x_1^*, x_2^*)$ is the optimal consumption bundle on B , where the indifference curve I_1 is tangent to the budget line. Since $x_1^* > \bar{x}_1$ and $\bar{x}_2 > x_2^*$ the consumer is maximizing utility by selling commodity 2, which gives her receipts of $p_2(\bar{x}_2 - x_2^*)$, and buying commodity 1 at a cost of $p_1(x_1^* - \bar{x}_1)$.

Increases in p_1 relative to p_2 will make the budget line rotate clockwise about $\bar{x}$ and the optimal bundle will vary as p_1/p_2 changes, as the upper half of Fig. 2.17 illustrates. With the budget line at B_2 the optimal bundle is the endowed bundle $\bar{x}$, and the consumer does not trade at all on the market. A further increase in p_1/p_2 will shift the budget line to B_3 where the optimal bundle is x' and the consumer is now selling commodity 1 and buying commodity 2.

Figure 2.17



The line FF in Fig. 2.17 is the locus of optimal bundles traced out as p_1/p_2 varies with $\bar{x}$ fixed and is called the *offer curve*, since it shows the amounts (positive or negative) of the two goods which the consumer offers on the market at different relative prices. The consumer's *consumption demand curve* DD in the lower half of Fig. 2.17, which plots the consumption of x_1 as a function of p_1/p_2 , is derived from the offer curve. As p_1 increases relative to p_2 the consumer reduces consumption of commodity 1, from x_1^* to $\bar{x}_1$ and then to x_1' as she moves along FF from x^* to $\bar{x}$ and x' . $(p_1/p_2)_1$, $(p_1/p_2)_2$ and $(p_1/p_2)_3$ are the price ratios at which x^* , $\bar{x}$ and x' are chosen.

The $\hat{D}\hat{D}$ curve in part (b) of Fig. 2.17 is the consumer's *net demand curve* and plots the net demand $\hat{x}_1 = x_1 - \bar{x}_1$, the amount of commodity 1 that she buys or sells on the market, against (p_1/p_2) . It is derived by taking the horizontal distance between the DD curve and a vertical line through $x_1 = \bar{x}_1$ at each price ratio. Notice that when $(p_1/p_2) > (p_1/p_2)_2$ the consumer's net demand is negative: she supplies commodity 1 to the market.

In the illustrations above the effect of a fall in the relative price of a commodity was to increase the consumer's demand for it. However, with a different indifference map the DD and $\hat{D}\hat{D}$ curves could have been positively sloped, indicating that rises in the relative price of commodity 1 reduce the amount of the commodity supplied to the market. Hence a *ceteris paribus* change in a single price may increase, decrease or leave unchanged the individual's consumption of any single commodity. Similarly, a *ceteris paribus* change in the initial endowment may increase, decrease or leave unchanged the consumption of commodity i .

The consumption decision in terms of net demands

The consumer's optimization problem studied earlier had the levels of consumption $(x_1, \dots, x_n)$ as choice variables, but it is possible to formulate the problem with the consumer's net demands $(\hat{x}_1, \dots, \hat{x}_n)$ as the choice variables. Since this particular approach will be used in Chapter 12 on general equilibrium, it is useful to show here that it is equivalent to the model of section C.

The consumer's utility function $u(x)$ can be rewritten as a function of the net demands $\hat{x}_i$ since from the definitions $\hat{x}_i \equiv x_i - \bar{x}_i$ we have $x_i \equiv \hat{x}_i + \bar{x}_i$ and so

$$u(x_1, \dots, x_n) = u(\hat{x}_1 + \bar{x}_1, \dots, \hat{x}_n + \bar{x}_n) \quad [\text{E.4}]$$

Since the initial endowments $\bar{x}_i$ are constants, u varies only as the $\hat{x}_i$ vary:

$$u(\hat{x}_1 + \bar{x}_1, \dots, \hat{x}_n + \bar{x}_n) = \hat{u}(\hat{x}_1, \dots, \hat{x}_n) \quad [\text{E.5}]$$

and

$$\frac{\partial \hat{u}}{\partial \hat{x}_i} = \frac{\partial u}{\partial \hat{x}_i} = \frac{\partial u}{\partial x_i} \frac{\partial x_i}{\partial \hat{x}_i} = \frac{\partial u}{\partial x_i} \quad [\text{E.6}]$$

$\hat{u}$ will have all the properties possessed by u such as continuity, quasi-concavity, etc.

The feasible set can also be rewritten in terms of the $\hat{x}_i$, as was shown in [E.2].

The non-negativity constraints on the x_i are replaced by

$$x_i = \hat{x}_i + \bar{x}_i \geq 0 \quad [\text{E.7}]$$

or

$$\hat{x}_i \geq -\bar{x}_i \quad i = 1, \dots, n$$

i.e. the supply of a good cannot exceed the endowment of that good. The consumer's optimization problem can now be written in terms of the net demands as (compare [C.1]):

$$\begin{aligned} \max \quad & \hat{u}(\hat{x}_1, \dots, \hat{x}_n) \\ \text{s.t.} \quad & \sum p_i \hat{x}_i \leq 0 \\ & \hat{x}_i \geq -\bar{x}_i \quad i = 1, \dots, n \end{aligned} \quad [\text{E.8}]$$

Proceeding, as in section C, to assume that the direct constraints [E.7] do not bind at the solution, the Lagrange function of the problem may be written

$$L = \hat{u}(\hat{x}_1, \dots, \hat{x}_n) - \lambda \sum p_i \hat{x}_i$$

First-order conditions are

$$\frac{\partial L}{\partial \hat{x}_i} = \hat{u}_i - \lambda p_i = 0 \quad i = 1, \dots, n \quad [\text{E.9}]$$

$$\frac{\partial L}{\partial \lambda} = -\sum p_i \hat{x}_i = 0 \quad [\text{E.10}]$$

and from [E.6] we see that [E.9] is identical to [C.7], so that we would be able to derive exactly the same equilibrium conditions and comparative static properties as in section C.

This reformulation of the problem in terms of net demands rather than consumption bundles is, in terms of the diagrammatic analysis of Fig. 2.17, equivalent to shifting the origin to $\bar{x}$ so that $\hat{x}_1 = x_1 - \bar{x}_1$ and $\hat{x}_2 = x_2 - \bar{x}_2$ are measured along the axes. The budget line now passes through the new origin and the consumer's indifference map is unaffected. The reader should redraw Fig. 2.15 and the upper part of 2.16 in this way to confirm that nothing of substance is affected by the relabelling.

EXERCISE 2E

1. Show that, when the consumer is a net seller of good x_1 , a fall in price of this good can result in a fall in demand for it, even given that it is a normal good. Explain intuitively why this is the case. Show that it does not happen if the consumer is a net buyer of the good.
2. How would you interpret the slope of the consumer's offer curve?
3. Explain why, at every point on the offer curve, there is tangency between an indifference curve and the budget line generating that point.
4. Discuss the relevance of the model examined in this section to:
 - (a) a market in stocks and shares;
 - (b) a market in new and secondhand cars;
 - (c) the case where x_1 is bread and x_2 is leisure time, with $\bar{x}_1 = 0$ and $\bar{x}_2 = 24$ hours per day.

Appendix 1: The lexicographic ordering

The lexicographic ordering shows the need for the continuity assumption if we wish to work with a numerical representation of the consumer's preference ordering, i.e. with a utility function. The lexicographic ordering can be shown to satisfy the first four assumptions set out in section A but to be incapable of being represented by a utility function. On the other hand, it can be shown to give rise to well-defined demand functions, which implies that the continuity assumption is not necessary for their existence.

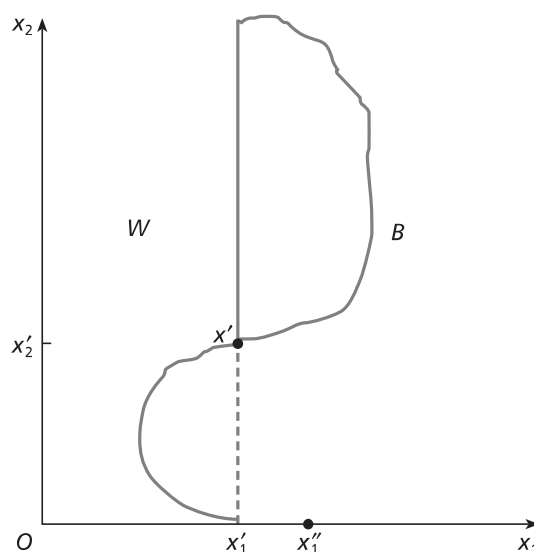
When consumption bundles consist of two goods, the ordering takes the following form. The consumer's preferences are such that, given two bundles $x' = (x'_1, x'_2)$ and $x'' = (x''_1, x''_2)$:

- (a) $x'_1 > x''_1$ implies $x' > x''$
- (b) $x'_1 = x''_1$ and $x'_2 > x''_2$ implies $x' > x''$

The consumer always prefers a bundle with more of the first good in it, regardless of the quantity of the second good; only if the bundles contain the same amount of the first good does the quantity of the second matter. An illustration would be the case of a drunkard who would always prefer a combination of beer and bread with more beer in it to one with less, regardless of the amount of bread, but if the amounts of beer are the same, prefers the one with more bread. It is called a 'lexicographic ordering' because it is analogous to the way words are ordered in a dictionary: words beginning with A always come before those beginning with B, but if two words begin with A then the second letter determines their order.

The indifference sets corresponding to this ordering are found with the help of Fig. 2.18. Take the bundle $x' = (x'_1, x'_2)$, and ask: what points are preferred to it, and to what points is it preferred? The area B , including the points on the solid line above

Figure 2.18



x' , must all be preferred to x' , since points to the right of x' contain more x_1 , while points along the solid line contain as much x_1 and more x_2 . The area W , including the points on the broken line below x' , must all be such that x' is preferred to them, since points to the left of x' contain less x_1 , while points on the broken line contain as much x_1 but less x_2 . But if all the points in B are preferred to x' , and x' is preferred to all the points in W , there can be no other points indifferent to x' , and so the indifference set for x' consists only of this single point. Since x' was chosen arbitrarily, this is true for every point in the space: each lies in an indifference set consisting only of itself.

The lexicographic ordering does not satisfy the assumption of continuity, since the indifference set is a point and not a continuous curve. If we reduce the amount of x_1 in the bundle, by however small an amount, we can find no amount of x_2 to compensate for the change (the drunkard cannot be bribed by any amount of bread to give up even a sip of beer). We now show that it is not possible to represent the lexicographic ordering by a utility function.

First, if we divide the real line into non-empty, disjoint bounded intervals, the set of these intervals is *countable*. That is, we can put them into a one-to-one correspondence with the set of positive integers $\{1, 2, 3, \dots\}$. On the other hand, the points on the real line itself or some interval of it, e.g. its positive half, are *not* countable. It follows that any argument which leads to the conclusion that the positive half of the real line is countable must be false. We can show that the assumption that a utility function exists for the lexicographic ordering does just that.

Suppose that a utility function $u(x_1, x_2)$ exists, which gives a numerical representation of the lexicographic ordering. Refer to Fig. 2.18. Setting $x_1 = x'_1$, the function will take on the values $u(x'_1, x_2)$ for all $x_2 \geq 0$ along the vertical line in the figure. This set of values has a lower bound at $u(x'_1, 0)$, and it must have an upper bound because the u -values for any $x_1 = x''_1 > x'_1$ must be greater. Hence this set of values represents a non-empty bounded interval on the real line. Now choosing a value $x''_1 > x'_1$ we can in the same way associate with it a non-empty bounded interval of real numbers. Moreover, however close to x'_1 we choose x''_1 , it must always be the case that the intervals are disjoint, since $u(x''_1, x_2) > u(x'_1, x_2)$ for every x_2 , since $x''_1 > x'_1$. We can repeat this argument for every value of x_1 on the horizontal axis: to each corresponds a unique non-empty bounded interval on the real line. But this means we have put the positive half of the real line into one-to-one correspondence with a set of non-empty disjoint intervals of the real line, implying that the former is countable. This is false, and therefore so is the initial assumption that the utility function exists.

As we suggested earlier, the fact that the lexicographic ordering does not possess a numerical representation does not mean the consumer's choice problem cannot be solved nor even that a continuous demand function does not exist. Remember that continuity assumptions are usually sufficient rather than necessary. Thus consider our beer drinker with lexicographic preferences, an income of M and facing price p_1 for beer and price p_2 for bread. He will *always* spend his entire income on beer and nothing on bread. Hence his demand functions are

$$x_1 = M/p_1, \quad x_2 = 0$$

which are well-defined and continuous. His demand curve for beer is just a rectangular hyperbola in (x_1, p_1) space and his demand function for bread in (x_2, p_2) space is the vertical axis.

EXERCISE A1

1. Suppose that the consumer has lexicographic preferences, but must consume a minimum level of x_2 for subsistence. Show how this affects his demand functions.
2. Likewise, show how the analysis is affected by the assumption that the consumer reaches a satiation level for x_1 , but not for x_2 .
3. Generalize the statement of the lexicographic ordering to n goods. What would be the demand functions with and without subsistence and satiation levels of each good?
4. How plausible do you find the assumption that a consumer has a lexicographic preference ordering with respect to:
 - (a) each good taken separately;
 - (b) groups of goods, e.g. food, clothing, shelter, entertainment?

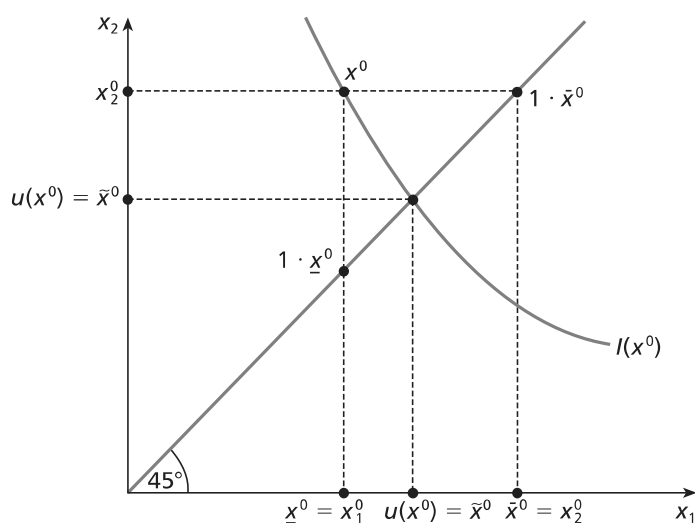
Appendix 2: Existence of a utility function

The lexicographic ordering satisfies completeness, reflexivity, transitivity and non-satiation, but no utility function can be constructed to represent it. We now show that adding the assumption of continuity guarantees that a continuous, increasing utility function can be found to represent the preference ordering. We do this by actually constructing such a function.

In Fig. 2.19, since the indifference curves are continuous, they intersect the 45° line as shown in the figure. For any point such as x^0 , associate with it the real number $u(x^0)$, a coordinate of the point at which the indifference curve through x^0 cuts the 45° line (either coordinate will do). The u -values are a numerical representation of this utility function: indifferent bundles have the same u -value, preferred bundles have higher u -values. We now put this more formally.

First, we need to state the continuity axiom in a more precise form. The better set of any point x^0 is $B(x^0) = \{x | x \succeq x^0\}$, and the worse set is $W(x^0) = \{x | x^0 \succeq x\}$. Then the continuity assumption is

Figure 2.19



For all bundles x^0 the sets $B(x^0)$ and $W(x^0)$ are closed.

The lexicographic ordering violates this assumption. Recall that a closed set is one which contains its boundary points, i.e. points having the property that every neighbourhood of them contains points which are, and points which are not, in the set. In Fig. 2.18, the boundary of $B(x')$ and $W(x')$ is the vertical line through x' . Points on the dotted line below x' are not in $B(x')$, points on the solid line above x' are not in $W(x')$, and so neither of these sets is closed. Loosely, this means that one can move from a point strictly preferred to x' , to a point strictly inferior to x' , without passing through a point indifferent to x' , however close to x' these points may be. In this loose sense there is a 'jump' in the preference ordering. The continuity assumption removes this possibility.

We now construct the utility function. First, we show that associated with any x^0 is a unique number $u(x^0)$, then that this has the properties of a utility function, and finally that this function is continuous.

Existence of $u(x^0)$: Given any bundle $x^0 = (x_1^0, \dots, x_n^0)$, choose the smallest and largest components, and denote them by $\underline{x}^0$ and $\bar{x}^0$ respectively. If $\mathbf{1}$ is the n -vector $(1, \dots, 1)$, then we have $\mathbf{1} \cdot \bar{x}^0 \geq x^0 \geq \mathbf{1} \cdot \underline{x}^0$ (Fig. 2.19 illustrates).

If $\bar{x}^0 = \underline{x}^0$ then the proposition holds trivially so we ignore this case. Consider the non-empty interval of real numbers $[\underline{x}^0, \bar{x}^0]$. We claim that there exists a number $\tilde{x}^0 \in [\underline{x}^0, \bar{x}^0]$ such that $\mathbf{1} \cdot \tilde{x}^0 \sim x^0$. Suppose not. Then for every $x \in [\underline{x}^0, \bar{x}^0]$ we have either $\mathbf{1} \cdot x > x^0$ or $x^0 > \mathbf{1} \cdot x$. Moreover, the transitivity and non-satiation assumptions imply that those x -values for which $\mathbf{1} \cdot x > x^0$ form a sub-interval of numbers strictly greater than the complementary sub-interval for which $x^0 > \mathbf{1} \cdot x$. This implies that the former sub-interval has a lower bound, and so a greatest lower bound; likewise the latter sub-interval has a least upper bound. Moreover, these bounds must be the same. Denote this common bound by b . Thus we can partition $[\underline{x}^0, \bar{x}^0]$ into $[\underline{x}^0, b]$ and $(b, \bar{x}^0]$, or $[\underline{x}^0, b)$ and $[b, \bar{x}^0]$. In each case, $\mathbf{1} \cdot b$ is a boundary point of $W(x^0)$ and $B(x^0)$, and is either not contained in $W(x^0)$ or not contained in $B(x^0)$. But this contradicts the assumption that these sets are closed. So there does exist $\tilde{x}^0$ such that $\mathbf{1} \cdot \tilde{x}^0 \sim x^0$, and we take $u(x^0) = \tilde{x}^0$. The non-satiation assumption implies that $u(x^0)$ is unique.

We now have to show that the $u(x)$ numbers constructed in this way satisfy the definition of a utility function, which is, for any two bundles x^0, x' ,

$$u(x^0) \geq u(x') \Leftrightarrow x^0 \geq x'$$

Proof

(a) $u(x^0) \geq u(x') \Rightarrow x^0 \geq x'$. Suppose not, i.e. $u(x^0) \geq u(x')$ but $x' > x^0$. Then $\mathbf{1} \cdot u(x^0) \geq \mathbf{1} \cdot u(x')$, where $\mathbf{1}$ is again an n -vector of ones. We then have by transitivity

$$\mathbf{1} \cdot u(x') \sim x' > x^0 \sim \mathbf{1} \cdot u(x^0)$$

which by non-satiation gives the contradiction

$$u(x') > u(x^0)$$

(b) $x^0 \geq x' \Rightarrow u(x^0) \geq u(x')$. Suppose not, i.e. $x^0 \geq x'$ but $u(x') > u(x^0)$. Then $\mathbf{1} \cdot u(x') > \mathbf{1} \cdot u(x^0)$ and the chain $x' \sim \mathbf{1} \cdot u(x') > \mathbf{1} \cdot u(x^0) \sim x^0$ gives the contradiction.

Finally, to prove that $u(x)$ is a continuous function it is convenient to take the following property of continuous functions.

A function $u(x)$, $x \in \mathbb{R}_+^n$, is continuous on $\mathbb{R}_+^n$ if and only if for each pair of subsets of function values, U_1 and U_2 , if U_1 and U_2 are separated then $u^{-1}(U_1)$ and $u^{-1}(U_2)$ are separated.

Two sets are separated if no point in one set is a boundary point of the other. Thus, for example, the pairs of sets $[0, 3/4]$ and $[1, 2]$, and $[0, 1)$ and $(1, 2]$ are separated, while the pair $[0, 1]$ and $(1, 2]$ are not.

Then, take any bundle x^0 and its corresponding utility value $u(x^0)$, and form the intervals $U_1 = [\underline{u}, u(x^0))$, $U_2 = (u(x^0), \bar{u}]$ where $\underline{u} < u(x^0)$ and $\bar{u} > u(x^0)$ are arbitrary. Then clearly U_1 and U_2 are separated. The set $u^{-1}(U_1) = \{x | u(x) \in U_1\}$ is a subset of the interior of $W(x^0)$, and the set $u^{-1}(U_2) = \{x | u(x) \in U_2\}$ is a subset of the interior of $B(x^0)$. Since these subsets lie on either side of $I(x^0)$, which belongs to neither of them, they are also separated. Since x^0 , $\underline{u}$ and $\bar{u}$ were arbitrary, the function $u(x)$ is continuous.

The above discussion of the existence of a continuous, increasing utility function used a specific construction and also made use of the non-satiation assumption. It is possible to drop the assumption and still prove existence of a continuous utility function, but it requires more advanced mathematical methods.